

ENBC321/BIOE658I: Homework #8

Due: Monday 04/20/26 by 5:00 PM, beginning of the class

Submit your homework to ELMS, including Python code and the dataset

Overall points: 50

Note: Late homework may be submitted up to 24 hours after the due date, but a 50% penalty will be applied.

Note: Please aim to write clean, well-organized code that is easy for others to understand. Be sure to include comments or explanations where needed. Points may be deducted if this is not done.

Note: Please name your codes using the following format: `problem1_fullname.ipynb`. Start with the problem number and followed by your full name, e.g., **`problem1_jimmyazarnoosh`**.

Note: Run all cells and ensure there are no syntax errors. All results must be displayed in the notebook. (Syntax errors = zero points).

Note: Using generative AI for learning purposes is allowed. However, directly copying or closely replicating AI-generated content is strictly prohibited and will result in a grade of ZERO. Repeated violations may lead to more severe consequences.

Note: Submit **a single zip file** containing the following:

- The dataset you used
- Python code files in ***.ipynb**
- Your answers to the questions in PDF format

Problem 1. Patient Clustering with PCA and K-Means (50 points)

A hospital is analyzing the health profiles of patients to identify subgroups with similar clinical features, which may correspond to different disease subtypes or treatment needs. As a data scientist working with the medical research team, your task is to analyze patient data using **Principal Component Analysis (PCA)** and **K-means clustering**. The dataset contains anonymized data for 120 patients with the following features:

- **Age:** Patient's age (in years)
- **BMI:** Body Mass Index
- **Blood Pressure:** Systolic blood pressure (mmHg)
- **Glucose Level:** Fasting blood glucose (mg/dL)
- **Insulin Level:** Fasting insulin level ($\mu\text{U/mL}$)

Tasks:

- Load the dataset from the provided CSV file: [patients_data.csv](#).
- Standardize all features using `StandardScaler` so that each feature has mean = 0 and standard deviation = 1.
- Reduce the dimensionality from **5** features to **2** principal components using the following two approaches:
 - **Sklearn PCA:** Implement PCA using `sklearn.decomposition`. Save this code as: `problem01_name_sklearn.py`. (10 points)
 - **Manual PCA:** Compute PCA manually using NumPy (using eigen decomposition). Save this code as: `problem01_name_manual.py`. (10 points)
- Plot the 2D data after PCA transformation (for both **Sklearn** and **manual** versions). (5 points)
- Apply **K-Means clustering** (with 3 clusters) on the 2D PCA-transformed data. (5 points)
- Plot the clustered data in the 2D PCA space.

Conceptual questions:

1. Why is it important to standardize the data before applying PCA or K-Means? (5 points)
2. What does it mean when we say PCA “reduces dimensionality”? What are we losing or preserving in the process? (5 points)
3. Why might PCA be useful before applying clustering algorithms like K-Means? (5 points)
4. How much of the variance is explained by the first two principal components? How would you decide if two components are enough? (5 points)